CEGE0024

Structures and Materials: Plastic Analysis and Finite Element Modelling

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Seventh session objectives

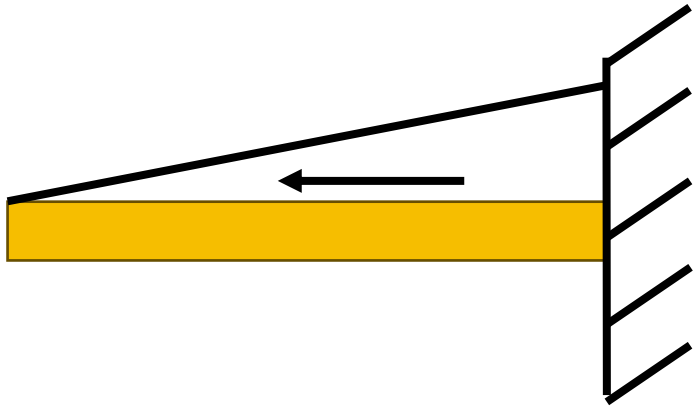
- Understand and be able to apply a mesh sensitivity analysis
- Understand basic checks you may complete to verify your model
- Be aware of different material models that are available

Recap: Sixth session objectives

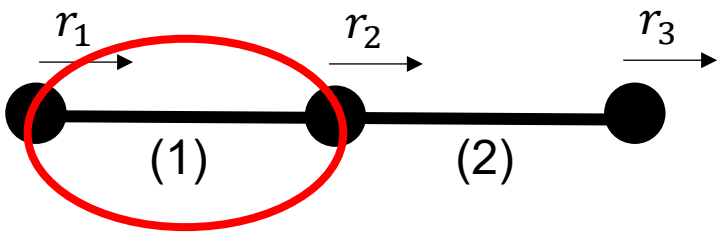
Brief introduction to FE modelling and understand the effect of the following **fundamental aspects** for applying an FE model:

- Discretisation
- Shape functions
- Element types

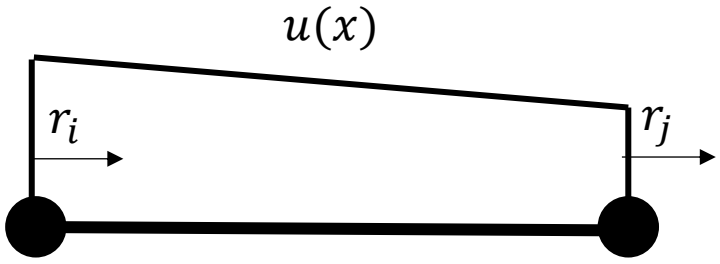
Recap from last week



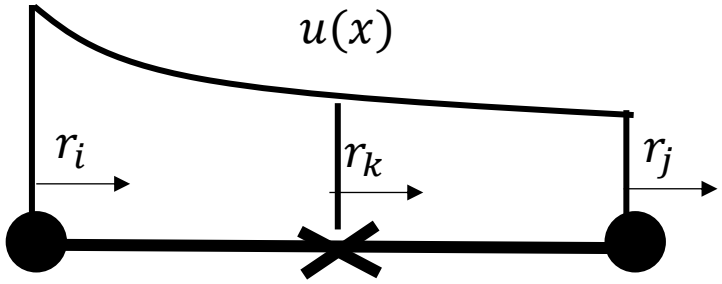
- Element type
- Level of discretisation



- Choice of shape function (element order)

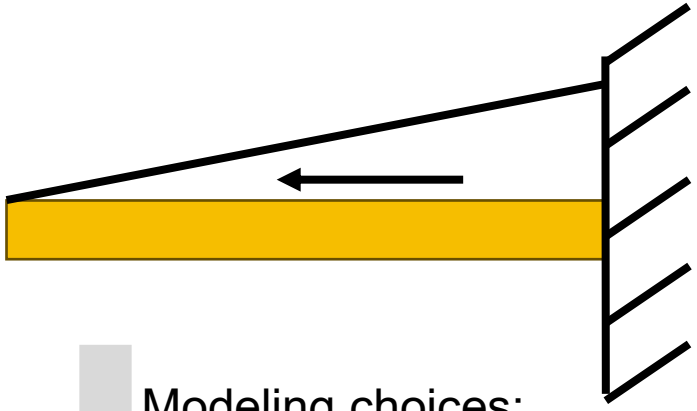


Linear

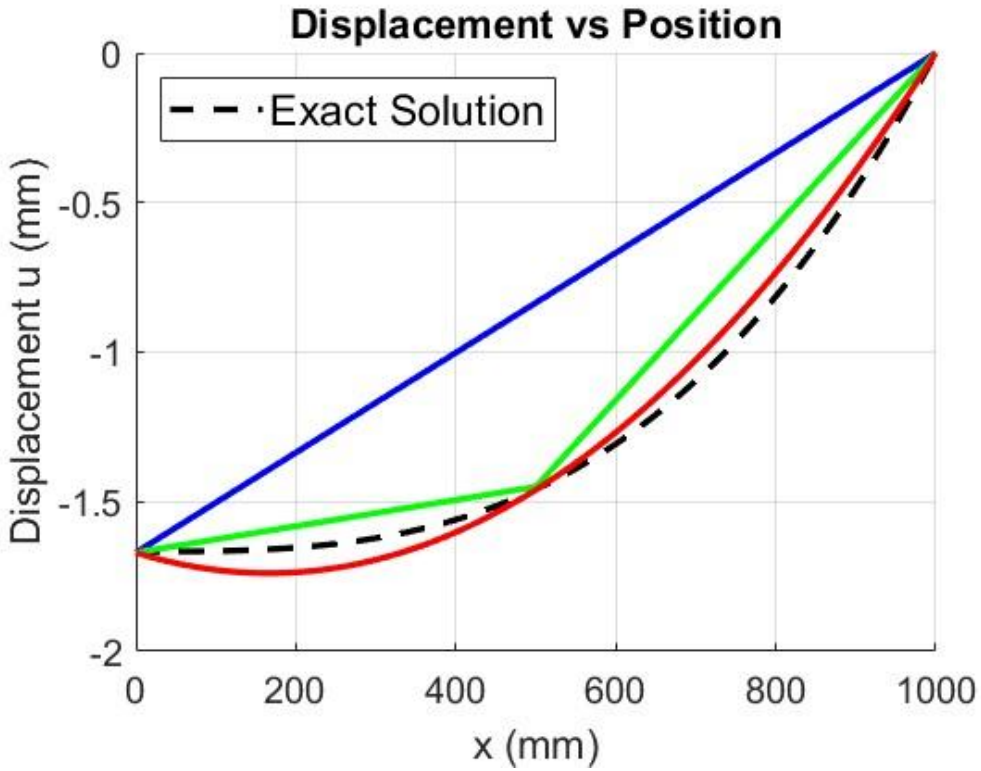


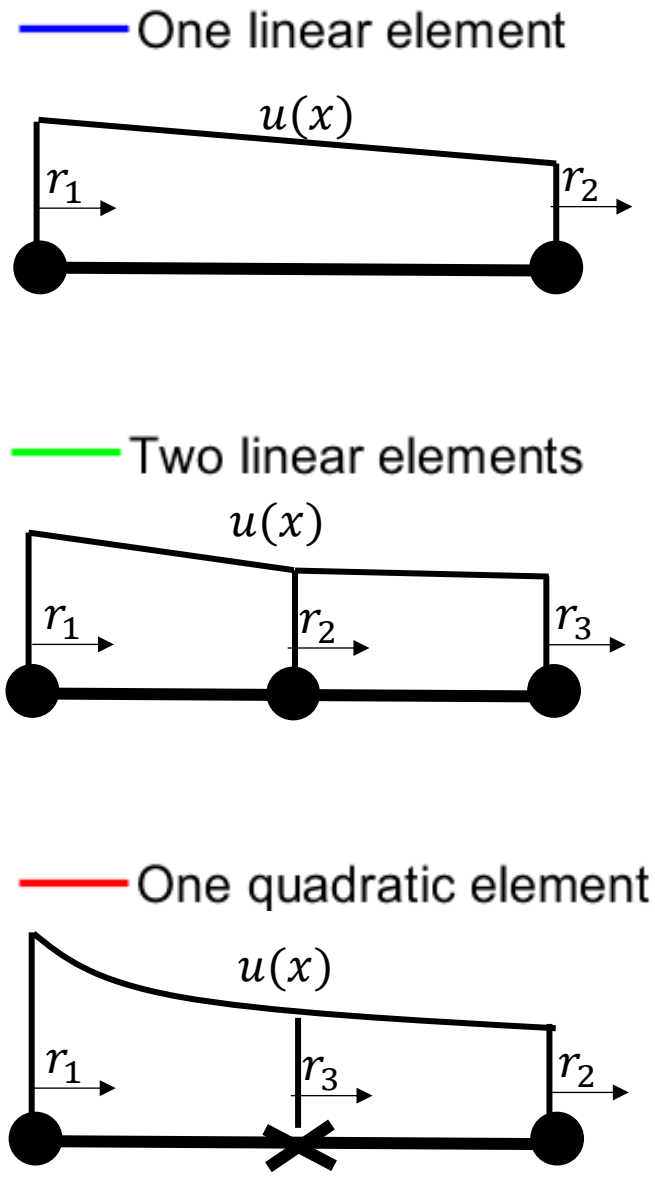
Quadratic

Recap from last week

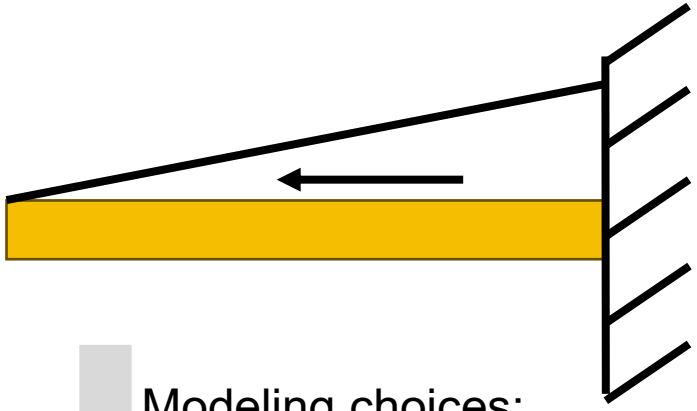


Modeling choices:
Element type
Level of discretization
Shape function

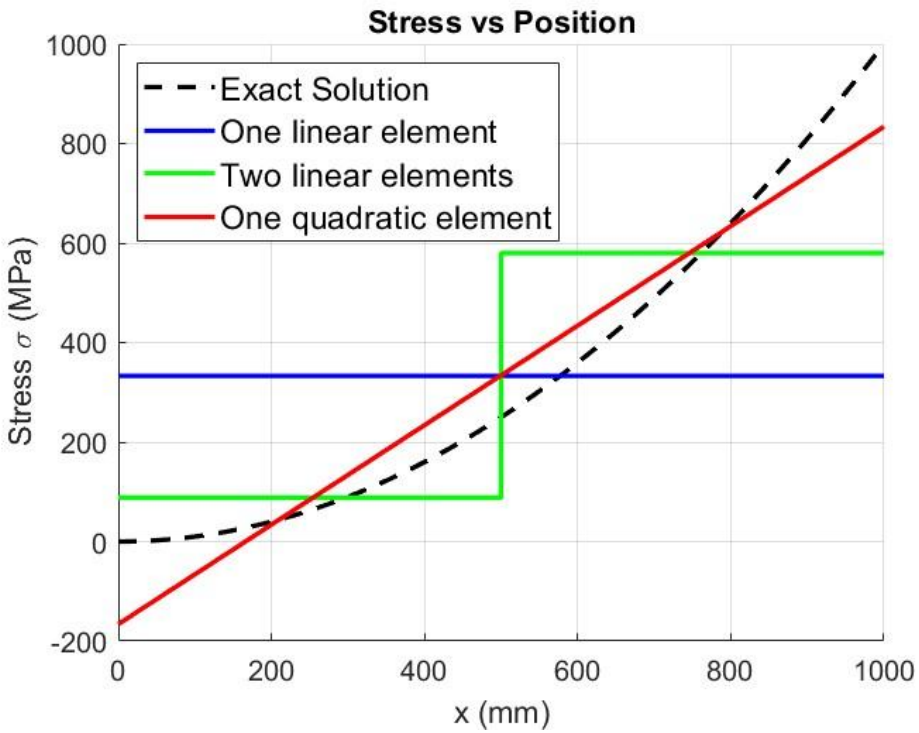




Recap from last week



<u>Modeling choices:</u>
Element type
Level of discretization
Shape function



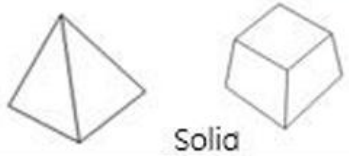
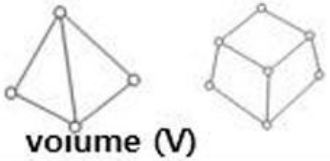
Constitutive equation

$$\sigma = E\varepsilon = E\frac{d}{dx}u(x)$$

- External equilibrium
- Internal equilibrium
  *It is approximate!

**In this example the exact displacement field is a cubic function, so to recover the exact solution we would need to assume a cubic shape function for the element. In this case internal equilibrium would be met.*

Recap from last week: Element types

Type	Actual Models	Finite Element Expressions (Geometric Properties Defined by Nodes)
1D	Rod (Truss) Beam	Length (L)
2D	Shell, Plane Stress, Plane Strain, Axisymmetric, etc	Area (A)
3D	 Solid	 volume (V)
Misc.	Spring, Mass, Rigid Link, etc.	-

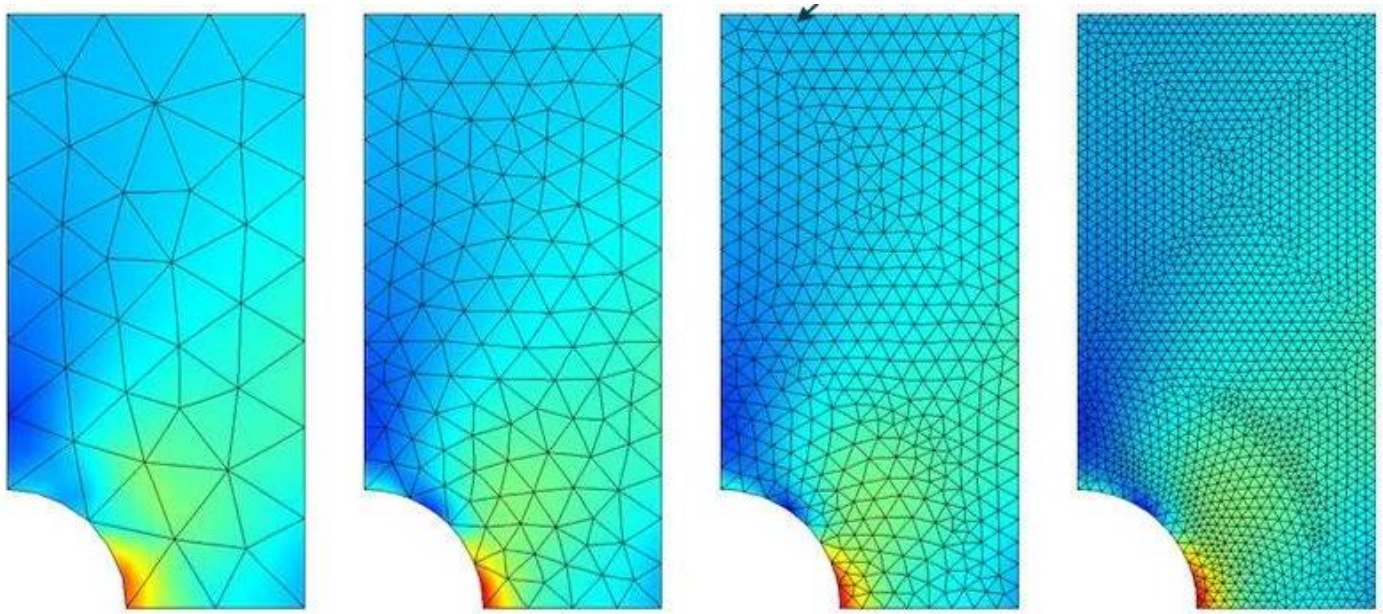
Reducing the dimension of the problem through assumptions

Use the lowest-dimensional element that still captures the physics you need!

[5] Sharma, K. (2019).

General points on discretisation

Coarse mesh → Fine mesh



How do we select an appropriate mesh density?

- A finer mesh density → generally increases accuracy
- Computationally → costs more
- Higher order shape functions → require less discretisation, and generally have less numerical issues

[6] Duffour, P (2026).

Mesh refinement

Discretisation error arises after the conceptual model is defined, when the structure is divided into finite elements.

This error comes from two main sources:

1. **Element behaviour error -** caused by the assumptions used to describe how each element deforms (the shape functions). Using higher-order shape functions generally gives a more accurate representation of the behaviour within each element.

2. **Mesh density error -** caused by how fine or coarse the mesh is. Increasing the number of elements (refining the mesh) usually improves accuracy and causes the solution to approach the exact solution of the conceptual model.

Finite element analysis gives an approximation, not the exact solution. Mesh refinement is how we check whether that approximation is **acceptable**.

[1] MacLeod, I. A. (2005)

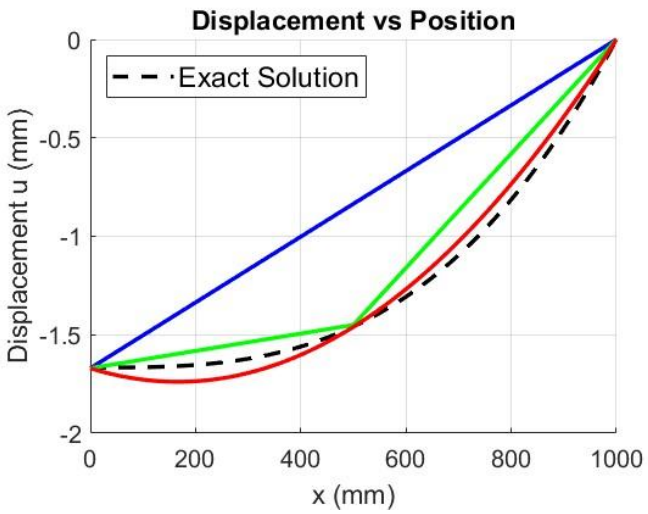
Mesh refinement

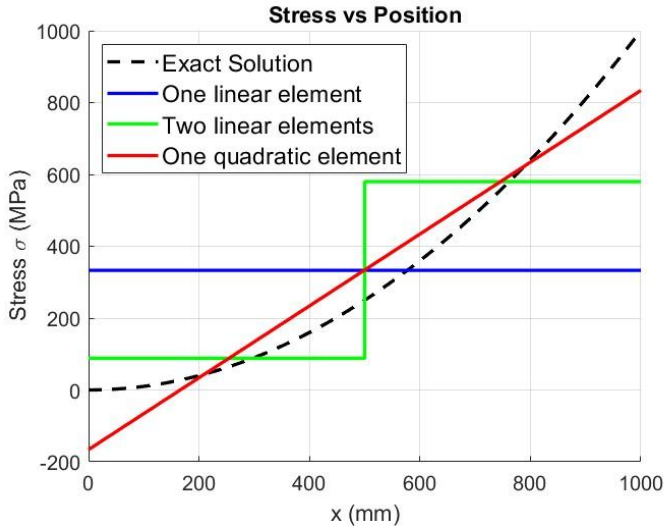
How do we determine whether that approximation is **acceptable?**

- If the results have convergence towards the exact solutions.

What results?

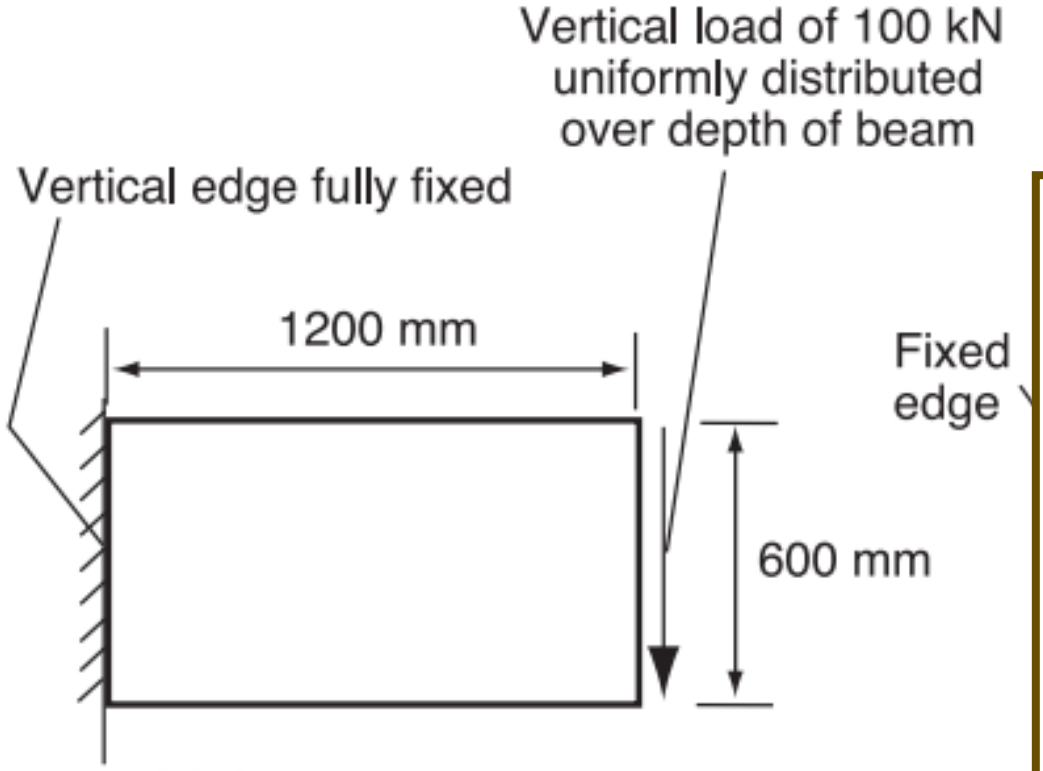
What if we do not know the exact solution?





Mesh refinement

How do we determine whether that approximation is **acceptable?**



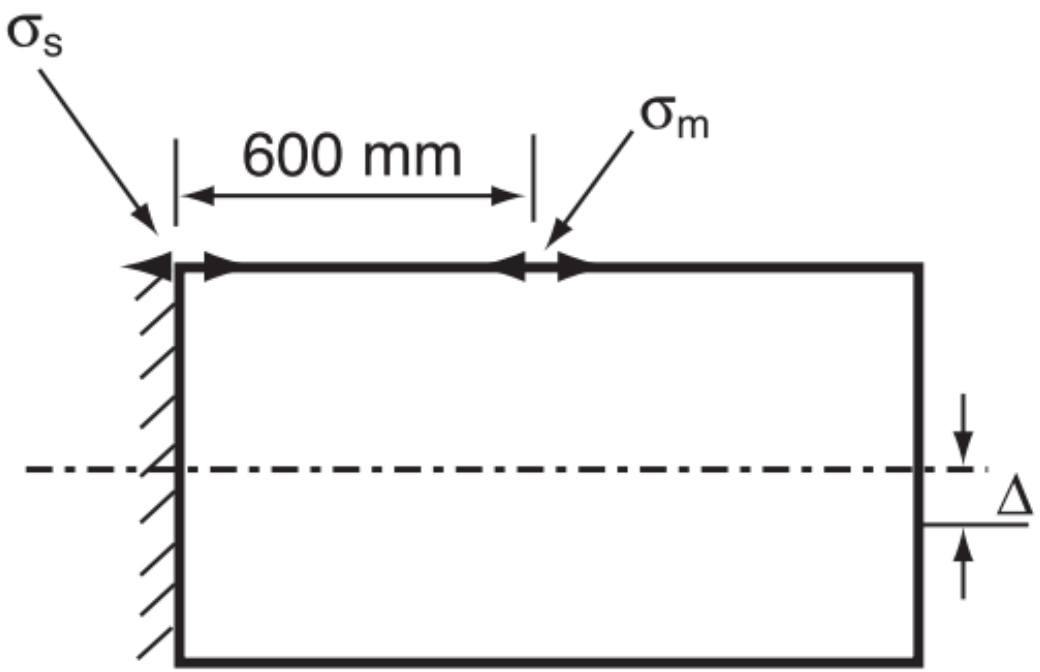
(a) Cantilever system

$E = 209\ \text{kN/mm}^2$, $\nu = 0.3$, thickness = 50 mm

[1] MacLeod, I. A. (2005)

Mesh refinement

How do we determine whether that approximation is **acceptable?**

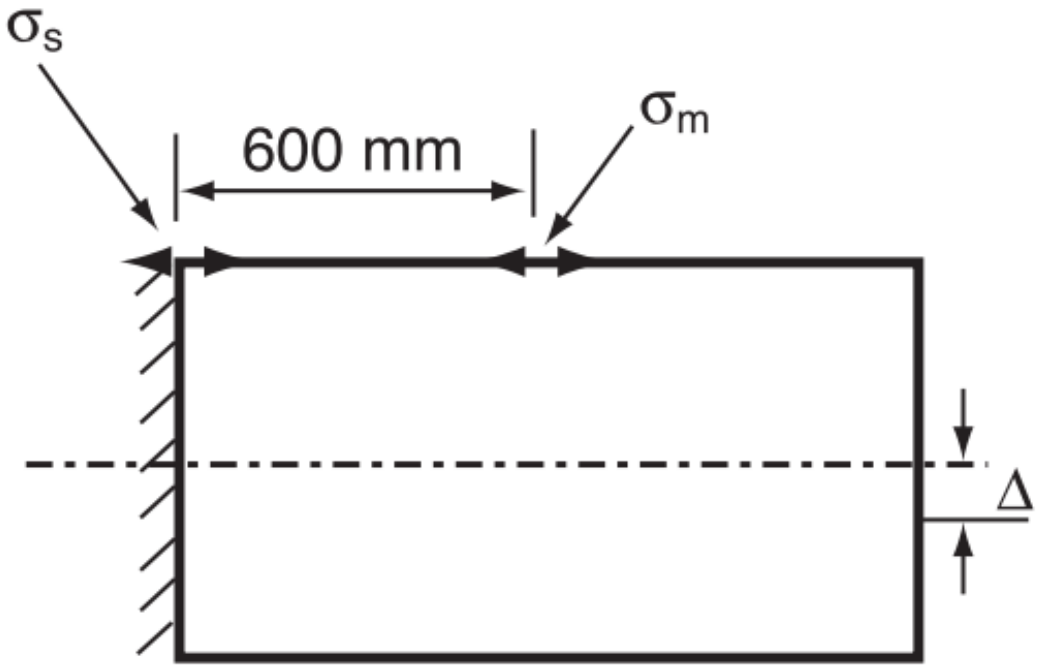


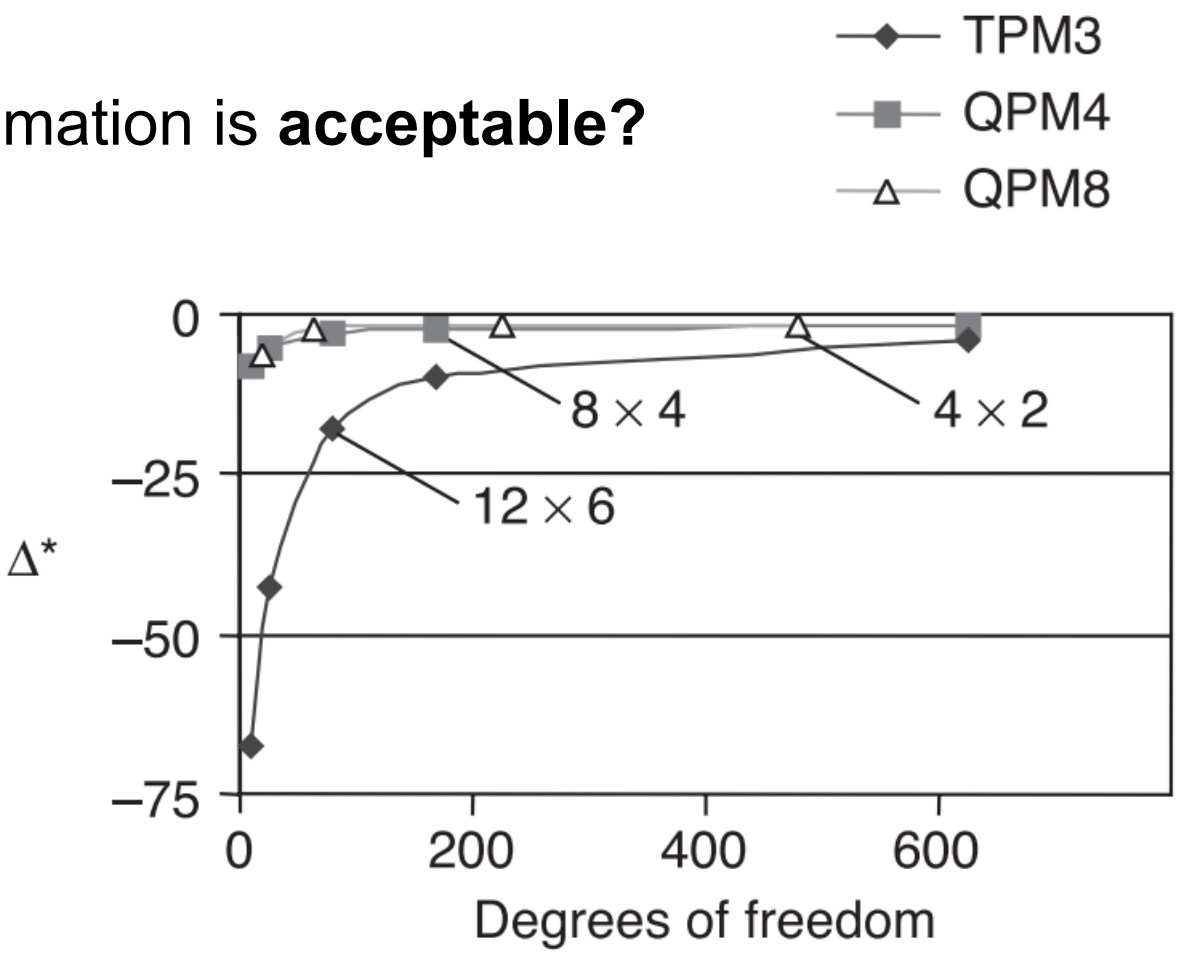
Mesh	*Dof*	
	TPM3, QPM4	QPM8
2 × 1	8	20
4 × 2	24	64
8 × 4	80	224
12 × 6	168	480
24 × 12	624	

[1] MacLeod, I. A. (2005)

Mesh refinement

How do we determine whether that approximation is **acceptable?**



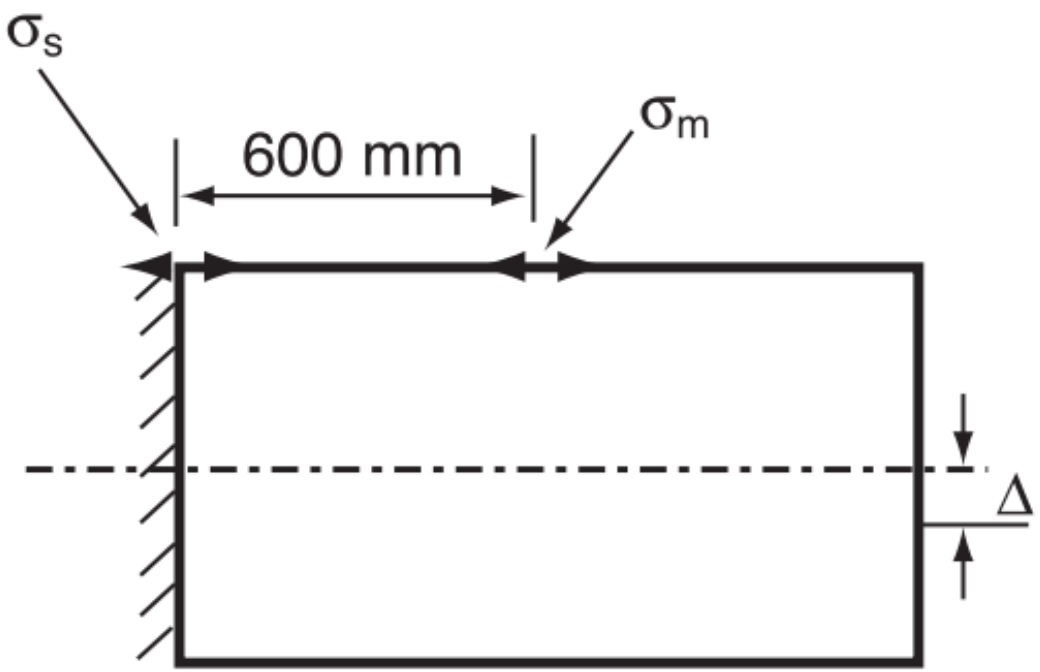


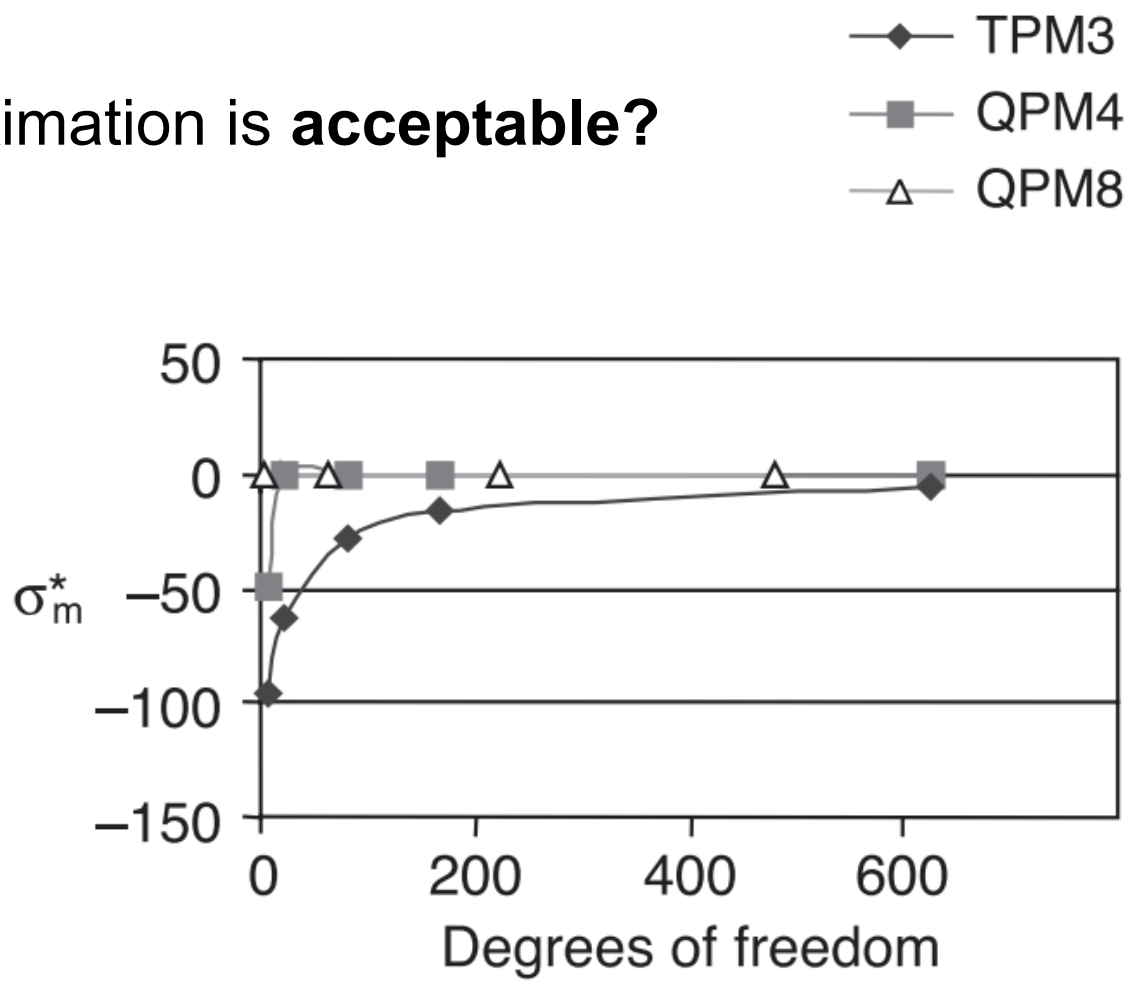
(a) Tip deflection (including mesh sizes)

[1] MacLeod, I. A. (2005)

Mesh refinement

How do we determine whether that approximation is **acceptable?**



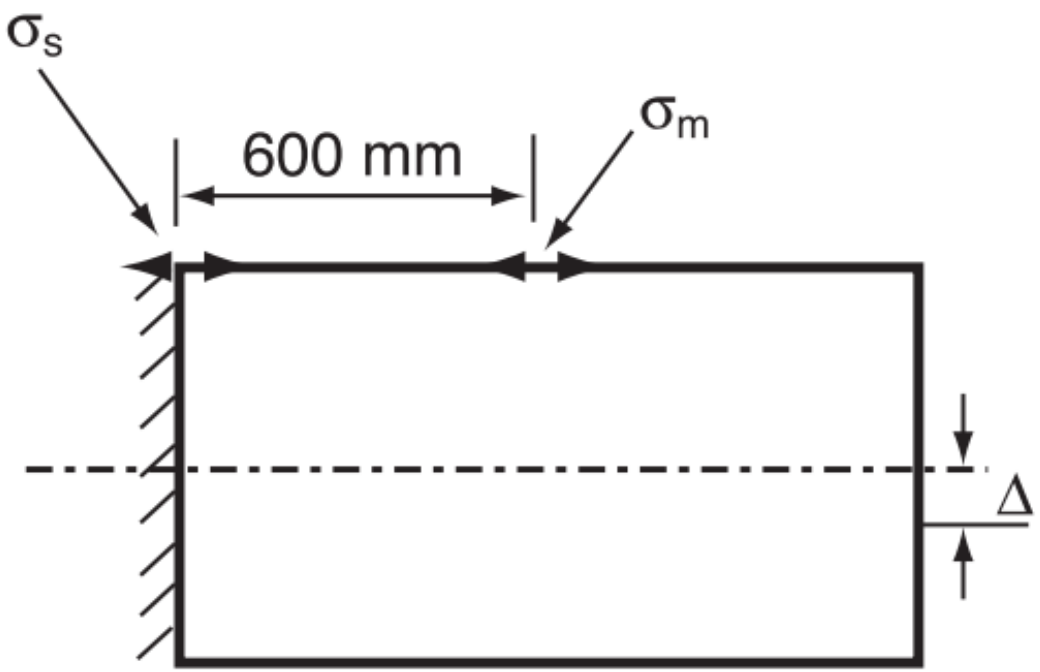


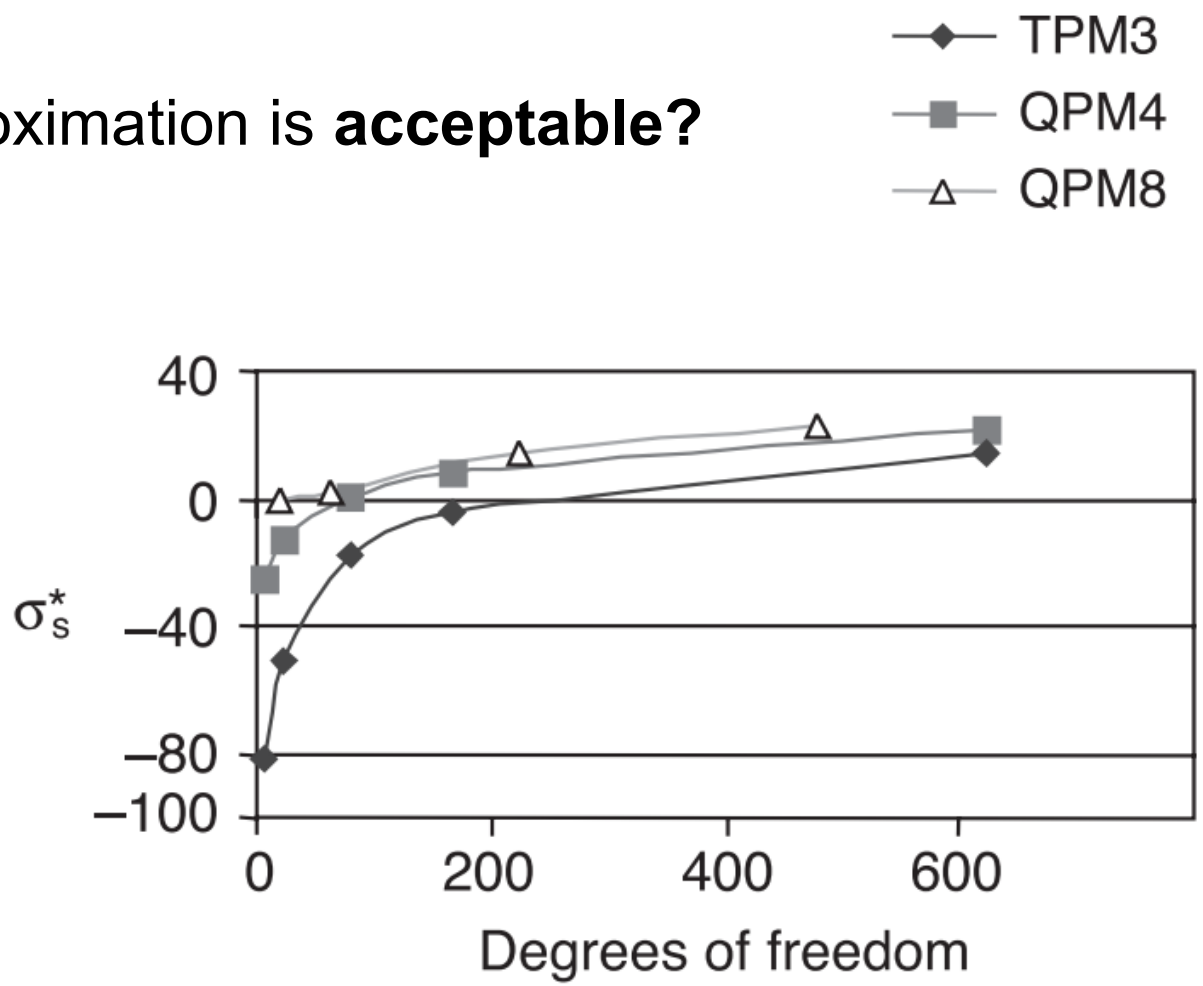
(b) Stress at midspan

[1] MacLeod, I. A. (2005)

Mesh refinement

How do we determine whether that approximation is **acceptable?**





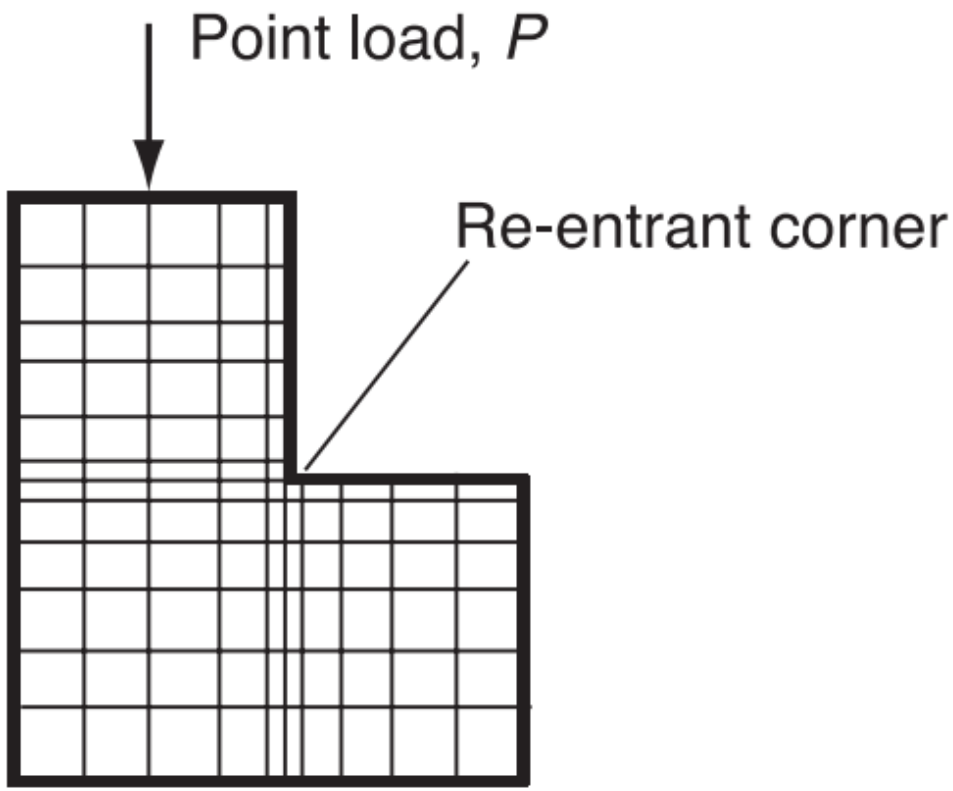
(c) Bending stress at support

We have a stress singularity here!

[1] MacLeod, I. A. (2005)

Mesh refinement

Singularities arise from idealised geometry and the material model. They are **not** a numerical artifact.



Types of singularities:

1. Re-entrant corners

Analytical (exact) solutions for linear elastic stress around a corner with radius *r*:

$$\sigma \propto r^{\lambda-1}$$

Where: λ depend on the geometry of the corner and is within the bounds $0 < \lambda < 1$

A sharp internal corner creates infinite stress!

[1] MacLeod, I. A. (2005)

Mesh refinement

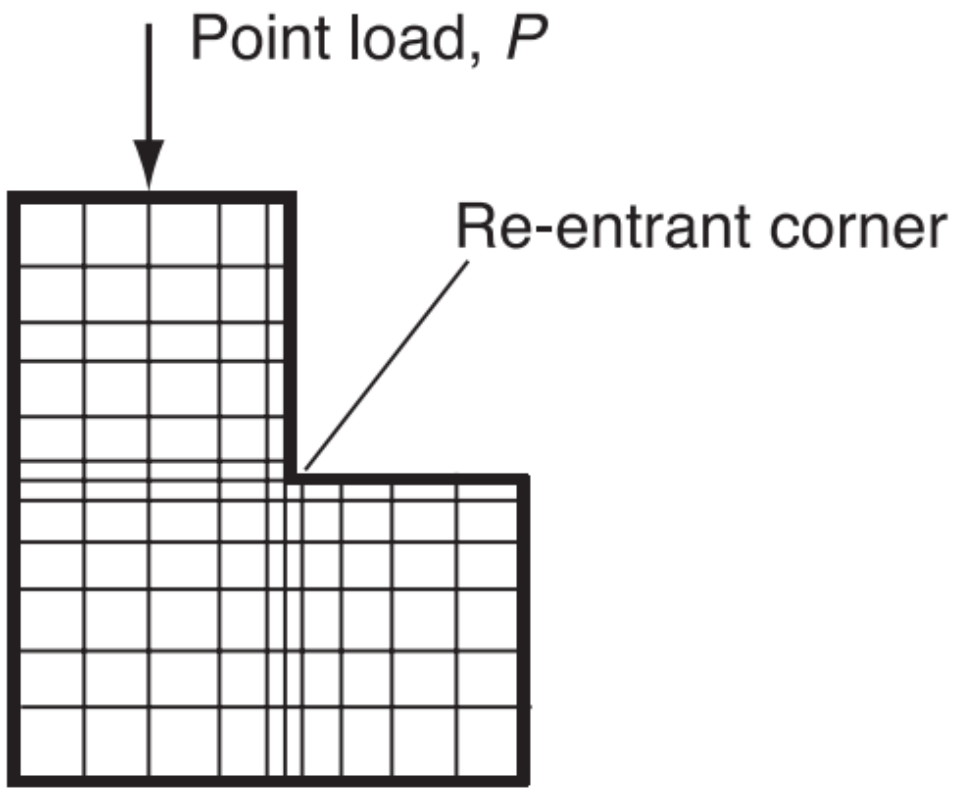
Singularities arise from idealised geometry and the material model. They are **not** a numerical artifact.

Types of singularities:

2. Point loads

Area of contact $A = 0$.

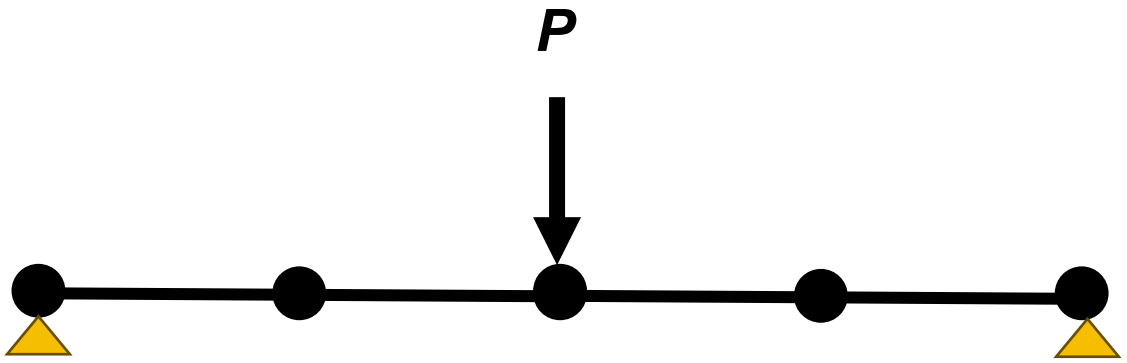
$$\sigma = \frac{P}{A} = \frac{P}{0} = \infty$$



[1] MacLeod, I. A. (2005)

Mesh refinement

- Can you have a singularity in 1D elements?



Mesh refinement

Should you apply modelling techniques to avoid singularities?

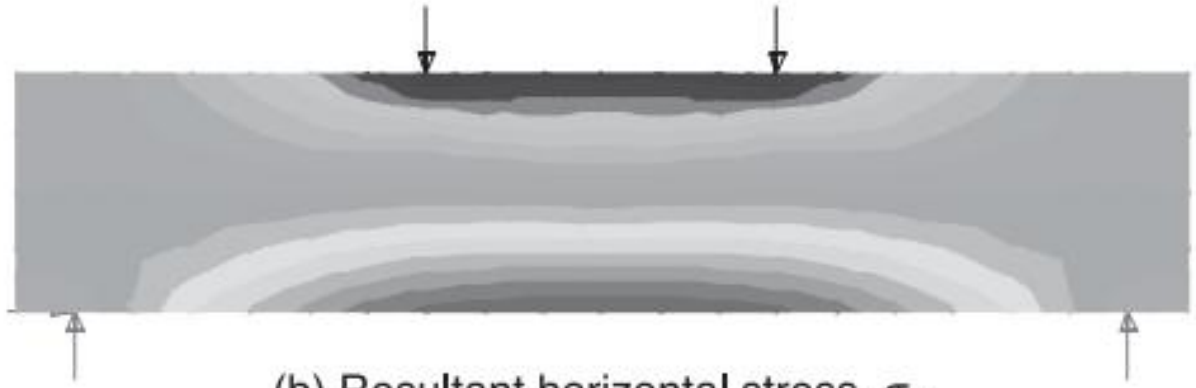
…..it depends what physics is important.

St-Venant principle: The detailed stress distribution near where a load is applied only matters locally. Far away, only the resultant force matter.

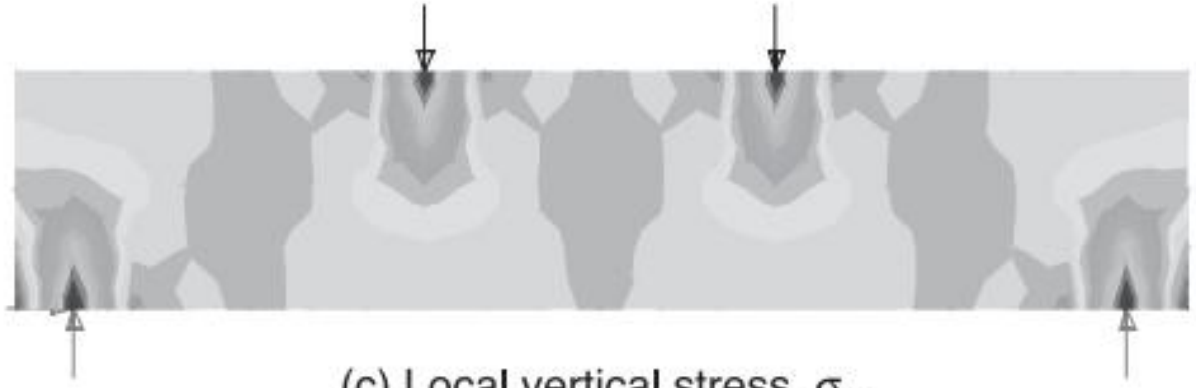
- Local stresses: highly dependent on how a load or constraint is applied
- Resultant stresses: govern global response (deformation, stresses)

Mesh refinement

St-Venant's principle: The detailed stress distribution near where a load is applied only matters locally. Far away, only the resultant force matter.



(b) Resultant horizontal stress, σ_x



(c) Local vertical stress, σ_y

"The St-Venant principle has the following consequences.

- The effect of local stresses tends to become negligible at distances twice the length over which the stress concentration acts.
- When calculating resultant stresses, the force actions can be replaced by static equivalents. This is not so for local stresses" [1]

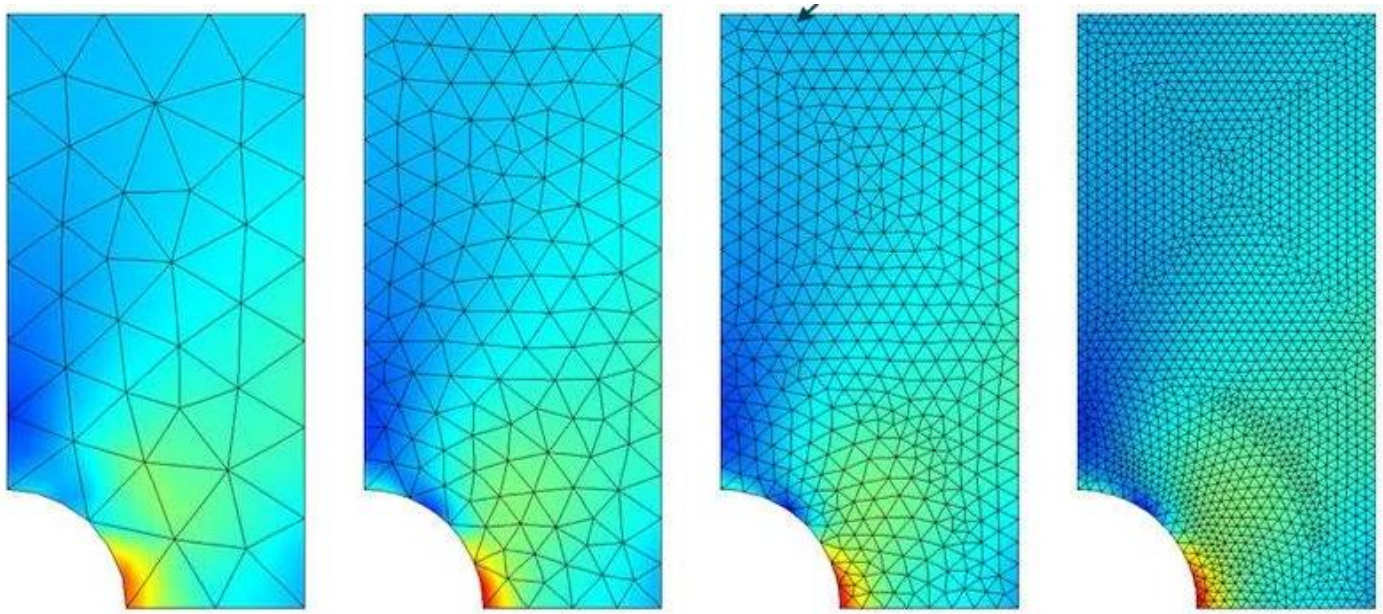
[1] MacLeod, I. A. (2005)

Mesh refinement

- Are the local stresses important for modelling the failure of this concrete beam?

Singularity vs stress concentration

Coarse mesh → Fine mesh



Do we need a fine mesh everywhere?

- **Mesh should be finer in areas where there are large gradients in the field variable.**

- Singularity $\sigma \rightarrow \infty$
- Stress concentrations are finite

[6] Duffour, P (2026).

Mesh refinement

How do we determine whether that approximation is **acceptable?**

- Convergence implies that as the mesh density is increased the results will converge towards the exact solution. The rate of convergence depends on the type of element and the number of elements in the mesh.

- A solution is converged when further refinement causes negligible change in the quantity of interest.

Mesh refinement

How do we determine whether that approximation is **acceptable?**

- You must decide what quantities you check for convergence!
  - Quantities of interest that should converge: Global displacements, reactions forces, internal stress (away from boundaries)
  - Quantities of interest that do not converge: peak stress at corners or loading points (singularities)

Mesh refinement

How to perform a basic mesh convergence study:

1. Choose a quantity of interest: i.e. δ

2. Create a refinement sequence: rule of thumb is to half the element size each iteration.

3. Plot or tabulate the result

4. Apply a convergence criterion: i.e. $\frac{|\delta_n - \delta_{n-1}|}{\delta_n} < 2\%$

Remember, a converged answer to an incorrect model is still wrong!

Model verification

"Do the results correspond to what is expected from the model? or have any errors been made in implementing the solution process?" [1]

Checklist:

- Error warnings in FE program outputs

- Overall equilibrium check

$$\sum F = R \approx 0$$

Where R are the residuals.

[1] MacLeod, I. A. (2005)

Model verification

"Do the results correspond to what is expected from the model? or have any errors been made in implementing the solution process?" [1]

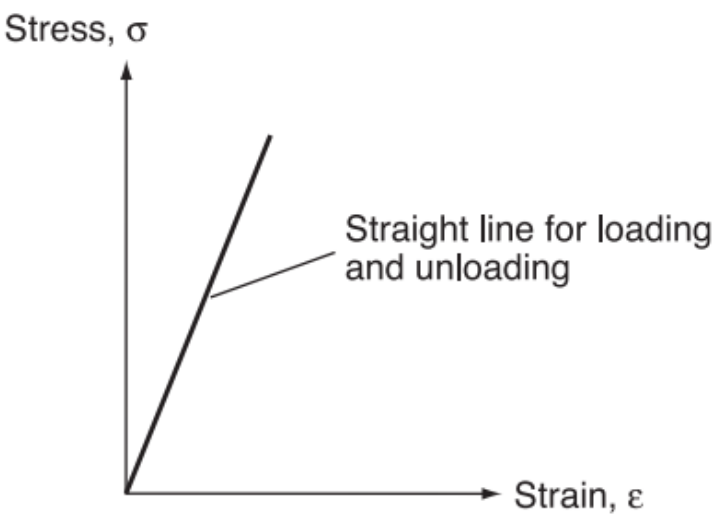
Checklist:

- Confirm restraints are correctly applied by inspecting the deformations here
- If the structure and loading is symmetric, check that the solution is also symmetric.
- Qualitative check: check the deflected shape, forces/stresses follow your expectation
- Quantitative check: simplified model/hand calculations to predict deformation or forces

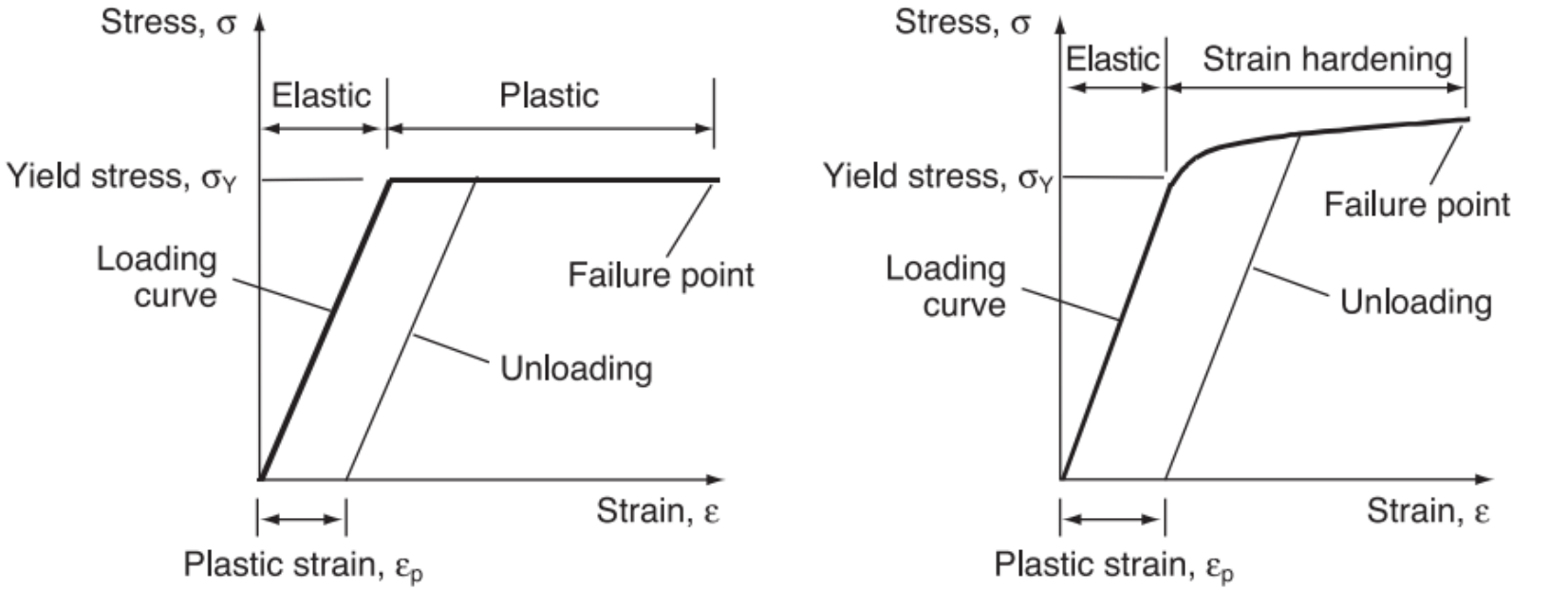
[1] MacLeod, I. A. (2005)

Material model

Liner elastic



Plastic



(a) Elastic-perfectly plastic

(b) Elastic-plastic with strain hardening

- Many other material models are available that may be required when trying to capture the correct physics of the problem.

[1] MacLeod, I. A. (2005)

References

[1] MacLeod, I. A. (2005). *Modern structural analysis : modelling process and guidance / Iain A. MacLeod.* Thomas Telford

[2] Debney, P., & Institution of Structural Engineers. (2020). *Computational engineering / Peter Debney.* Institution of Structural Engineers.

[3] Khennane, A. (2013). *Introduction to finite element analysis using MATLAB and Abaqus / Amar Khennane.* Taylor & Francis.

[4] Newman, G. M., J. T. Robinson, and C. G. Bailey. (2000). *Fire Safe Design: A New Approach to Multi-Storey Steel-Framed Buildings*. SCI Publication Fire and Steel Construction 288. Steel Construction Institute.

[5] Sharma, K. (2019). *Introduction to FEA* [SlideShare slides]. SlideShare. https://www.slideshare.net/KishanSharma26/introduction-to-fea-77358694

[6] Duffour, P. Personal communications – CEGE0009 lecture slides, (2026).

[7] Mandal, N. K., & Lees, M. (2016). *An investigation into monitoring rail stress in continuously welded rails through stress-free temperature* [Conference paper]. Conference on Railway Engineering, Melbourne, Australia.